

PHYSICS EXAMINATION 2024

Time: 2 Hours Max Marks: 50

SECTION A: Multiple Choice Questions (2 marks each)

1. Which of the following best describes Newton's First Law of Motion?
 - A. An object at rest stays at rest unless acted upon by an external force
 - B. Force equals mass times acceleration
 - C. For every action there is an equal and opposite reaction
 - D. Energy can neither be created nor destroyed

2. A ball is thrown vertically upward with initial velocity of 20 m/s.
Calculate the maximum height reached by the ball. ($g = 10 \text{ m/s}^2$)
 - A. 10 m B. 20 m C. 40 m D. 5 m

3. The speed of light in vacuum is approximately $3 \times 10^8 \text{ m/s}$. True or False?
 - A. True B. False

SECTION B: Short Answer Questions (5 marks each)

4. Refer to Figure 4.1 (circuit diagram below).

If $R_1 = 4 \text{ ohm}$, $R_2 = 6 \text{ ohm}$, $R_3 = 12 \text{ ohm}$ (parallel), find total resistance.

5. The velocity of a particle is recorded in the table below.

Time (s): 0, 2, 4, 6, 8

Velocity (m/s): 0, 10, 20, 30, 40

Draw a velocity-time graph and calculate the acceleration.

(Question continues on next page)

(Continued from Question 5)

Using the graph, determine:

- (a) Total displacement in 8 seconds
- (b) Average velocity

SECTION C: Long Answer (10 marks)

6. Explain the Law of Conservation of Energy with two real-world examples.
Derive the relationship between kinetic and potential energy for a freely falling object.

ANSWER KEY

1. A 2. B 3. A (True) 4. 2 ohm 5. 5 m/s^2